

Question 22

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Assume that you want to implement a new layer for a deep neural network that employs the following transformation

$$\mathbf{h}^{(l+1)} = \underbrace{\mathbf{M} \text{softmax}(\beta \mathbf{M}^T \mathbf{h}^{(l)})}_{:= \mathbf{p}},$$

where $\mathbf{h}^{(l)} \in \mathbb{R}^d$ is the input to this layer l , β is a positive hyperparameter to be set by the user, $\mathbf{M} \in \mathbb{R}^{d \times J}$ is a matrix of J patterns with d features provided by the domain expert, and $\mathbf{p}$ is defined as $\mathbf{p} := \text{softmax}(\beta \mathbf{M}^T \mathbf{h}^{(l)})$. Which of the following statements about this layer are true?

- ☒ a. $\mathbf{h}^{(l+1)}$ is equivariant to permutations of the columns of $\mathbf{M}$.
- ☒ b. $\mathbf{p}$ is invariant to permutations of the columns of $\mathbf{M}$.
- ☐ c. $\mathbf{p}$ is invariant to permutations of the columns of $\mathbf{M}$.
- ☐ d. $\mathbf{p}$ is equivariant to permutations of the columns of $\mathbf{M}$.
- ☒ e. $\mathbf{p}$ is equivariant to permutations of the rows of $\mathbf{M}$.
- ☐ f. $\mathbf{h}^{(l+1)}$ is invariant to permutations of the columns of $\mathbf{M}$.
- ☐ g. This layer cannot be built in deep learning architectures because gradients, i.e. $\frac{\partial \mathbf{h}^{(l+1)}}{\partial \mathbf{h}^{(l)}}$, cannot be backpropagated through it.

Question 21

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The main difference between the frequentist and Bayesian approach is marginalization over parameters when we define the probability for a class label given the input $\mathbf{x}$ and dataset $\mathcal{D}$. To give a more formal definition:

$$p(\mathbf{y}^{test} | \mathbf{x}^{test}, \mathcal{D}) = \mathbb{E}_{\mathbf{w} \sim p(\mathbf{w} | \mathcal{D})} [p(\mathbf{y}^{test} | \mathbf{w}, \mathbf{x}^{test})]$$

Deep Ensembles try to approximate the full expectation with (weighted) averages. Which of this options are valid?

$$\mathbb{E}_{\mathbf{w} \sim p(\mathbf{w} | \mathcal{D})} [p(\mathbf{y}^{test} | \mathbf{w}, \mathbf{x}^{test})] \approx$$

- ☒ a. $\approx \frac{1}{M} \sum_{m=1}^M p(\mathbf{y}^{test} | \mathbf{w}_m, \mathbf{x}^{test})$, where $\mathbf{w}_m$ denotes the weights of the m -th network of the ensemble.
- ☐ b. $\approx p(\mathbf{y}^{test} | \mathbf{w}, \mathbf{x}^{test})$
- ☒ c. $\approx \frac{1}{M} \sum_{m=1}^M \alpha_m p(\mathbf{y}^{test} | \mathbf{w}_m, \mathbf{x}^{test})$ with $\alpha_m = \text{softmax}(\mathbf{z})_m$ for some random normal distributed vector $\mathbf{z}$
- ☒ d. $\approx \frac{1}{M} \sum_{m=1}^M p(\mathbf{y}_m^{test} | \mathbf{w}, \mathbf{x}_m^{test})$ where the subscript m of the test data denotes the m -th equally sized fraction of the full data.
- ☐ e. $\approx \sum_{m=1}^M p(\mathbf{y}_m^{test} | \mathbf{w}, \mathbf{x}_m^{test})$ where the subscript m of the test data denotes the m -th equally sized fraction of the full data.
- ☐ f. $\approx \sum_{m=1}^M p(\mathbf{y}^{test} | \mathbf{w}_m, \mathbf{x}^{test})$

Question 15

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Suppose that you have a fixed, multi-dimensional, non-Gaussian distribution $p(\mathbf{x})$ that you want to approximate with a Gaussian distribution. You decide that you want to approximate this distribution with a Gaussian $q(\mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with some mean vector $\boldsymbol{\mu}$ and some covariance matrix $\boldsymbol{\Sigma}$. You now decide to minimize the Kullback-Leibler divergence between the two distributions $\text{KL}(p \parallel q)$ with respect to $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$.

Which expressions for $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ minimize the KL-divergence between the two distributions?

(The usual notation of the lecture is used: $\mathbb{E}$ is expectation, Cov is the covariance, $\mathbf{I}$ is the identity matrix)

- ☐ a. $\boldsymbol{\mu} = \mathbb{E}(\mathbf{x}^T \mathbf{x})$ and $\boldsymbol{\Sigma} = \mathbf{I}$
- ☒ b. $\boldsymbol{\mu} = \mathbb{E}(\mathbf{x})$ and $\boldsymbol{\Sigma} = \text{Cov}(\mathbf{x})$
- ☐ c. $\boldsymbol{\mu} = \mathbb{E}(\mathbf{x}^T \mathbf{x})$ and $\boldsymbol{\Sigma} = \mathbb{E}(\mathbf{x}^T)$
- ☐ d. $\boldsymbol{\mu} = \mathbb{E}(\mathbf{x})$ and $\boldsymbol{\Sigma} = \mathbf{I}$
- ☒ e. $\boldsymbol{\mu} = \mathbb{E}(\mathbf{x})$ and $\boldsymbol{\Sigma} = \mathbb{E}(\mathbf{x}\mathbf{x}^T)$

Question 2

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Consider the model defined in the question before with parameters initialized with zeros. $\mathbf{W}^{[2]}$ denotes the weight matrix of the second layer. You forward propagate a batch of examples, and then backpropagate the gradients and update the parameters. Which of the following statements is true?

Select one:

- ☒ a. The entries of the weight matrix $\mathbf{W}^{[2]}$ are all positive.
- ☐ b. The entries of the weight matrix $\mathbf{W}^{[2]}$ may be positive or negative.
- ☐ c. The entries of the weight matrix $\mathbf{W}^{[2]}$ are all negative.
- ☐ d. The entries of the weight matrix $\mathbf{W}^{[2]}$ are all zeros.

[Clear my choice](#)

Question 5

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You are given an input $\mathbf{x} = \begin{pmatrix} 1 & -2 \\ -1 & 2 \end{pmatrix}$ and a kernel weight matrix $\mathbf{w} = \begin{pmatrix} 0 & 1 \\ 2 & 3 \end{pmatrix}$. What is the result of the **transpose convolution operation** $*^T$ (stride 2) of

$$\mathbf{w} *^T \mathbf{x} = ?$$

Select one:

- ☐ a.
$$= \begin{pmatrix} 0 & 1 & -1 \\ 2 & -1 & -1 \\ -4 & -2 & 6 \end{pmatrix}$$
- ☐ b.
$$= \begin{pmatrix} 3 & 2 \\ 1 & 0 \end{pmatrix}$$
- ☒ c.
$$= \begin{pmatrix} 0 & 1 & 0 & -2 \\ 2 & 3 & -4 & -6 \\ 0 & -1 & 0 & 2 \\ -2 & -2 & 4 & 6 \end{pmatrix}$$
- ☐ d.
$$= \begin{pmatrix} 0 & 2 & 1 \\ 6 & -14 & 6 \\ 2 & 3 & 0 \end{pmatrix}$$
- ☐ e.
$$= \begin{pmatrix} 1 & 1 & 2 & 2 \\ 1 & 1 & -2 & -2 \\ 2 & 2 & 4 & 4 \\ 2 & 2 & -4 & -4 \end{pmatrix}$$
- ☐ f.
$$= (3)$$
- ☐ g.
$$= (42)$$
- ☐ h.
$$= \begin{pmatrix} 8 & -5 & 2 \\ -11 & 5 & 13 \\ 19 & 21 & -29 \end{pmatrix}$$

Clear my choice

Question 1

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You design a standard 2-layer fully connected neural network. All activations are sigmoids and your optimizer is stochastic gradient descent. You initialize all the weights and biases to zero and forward propagate an input $\mathbf{x}$ in the network. What is the output $\hat{y}$ of the network?

Select one:

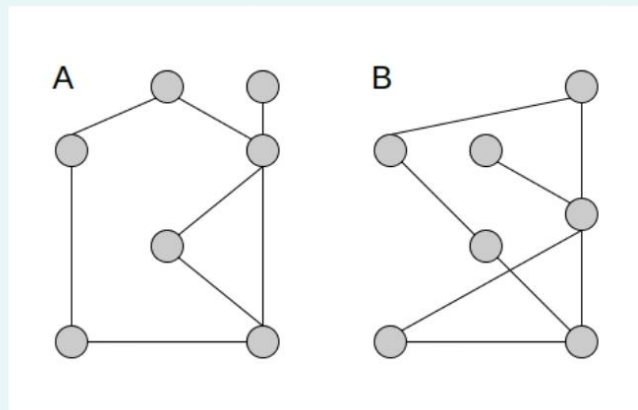
- ☐ a. 0
- ☒ b. $e^{\mathbf{x}}$
- ☐ c. 0.5
- ☐ d. -1
- ☐ e. 1

[Clear my choice](#)**Question 23**

Answer saved

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3.00 Flag question

Do the following two graphs fulfill the necessary condition for graph isomorphism, the Weisfeiler-Lehman test?



Select one:

- ☒ True
- ☐ False